FOR EDITH · OBSTETRICS & GYNAECOLOGY JUNIOR CLERKSHIP

Caesarean Section Instruments, Sutures & Aseptic Technique

Clinical skill | Week 3 · Medical disorders & PPH | Dr Ndiwalana

To assist safely at a caesarean section you must be able to name every instrument in the CS set and say what it is for, choose the right suture for each layer, and scrub, gown, glove, prepare and drape without contaminating the field. The count must be correct before anyone closes.

Must know before the ward round

1. Doyen retractor protects and retracts the bladder. Green-Armytage forceps grip the bleeding edges and angles of the uterine incision. Kocher's (toothed, heavy) and Allis (fine interlocking teeth) grip tough tissue such as the rectus sheath. Babcock (atraumatic, fenestrated) holds tubular structures such as the fallopian tube.
2. Mayo scissors cut heavy tissue (sheath). Metzenbaum scissors are long and fine for delicate dissection (peritoneum, bladder flap). Straight Mayo scissors cut sutures. Never cut sutures with Metzenbaum.
3. Suture classification uses 3 axes. Absorbable vs non-absorbable. Monofilament (smooth, less infection, more memory, needs more throws) vs braided (handles and knots well, but wicks bacteria). Natural (catgut, silk) vs synthetic (Vicryl, PDS, Monocryl, nylon, Prolene).
4. Gauge (USP) works backwards, so more zeros means thinner. Use
5. Use a round-bodied (taper) needle for soft tissue that parts easily (uterus, peritoneum, bowel, bladder). Use a cutting or reverse-cutting needle for skin and tough tissue.
6. WHO surgical hand preparation takes 2–5 minutes with antiseptic soap (chlorhexidine 4% or povidone-iodine 7.5%), or an alcohol-based surgical rub. Remove jewellery, keep nails short, no brushes on the skin, and keep hands above the elbows.
7. Skin prep is chlorhexidine-alcohol (WHO first choice) or povidone-iodine, applied from the incision outwards and allowed to DRY (fire risk with diathermy). Do not shave; clip if needed.
8. Count swabs, needles and instruments aloud (scrub nurse and circulator) before starting, before closing the uterus, before closing the peritoneum or sheath, and at skin closure. Never close on an incorrect count.
9. Check every sterile pack before opening. The indicator tape has changed colour, the pack is dry, intact and in date, and it has not been dropped. Wet, torn or unchanged means NOT sterile.
10. At the WHO Time out, confirm the patient, procedure and CS category, and that the antibiotic was given within 60 minutes before incision. Sign out confirms counts, specimens and concerns.
11. Incorrect count means stop closing, recount, explore the cavity, search the floor and bins, and X-ray (radio-opaque markers) before she leaves theatre if still missing. Document it.
12. Wound review. Pus or spreading redness on day 4–7 means SSI (swab before antibiotics, open and drain). Pink serous discharge on day 5–10 warns of fascial dehiscence (burst abdomen), which needs re-closure in theatre.

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1. Purpose and Indications

Most of what a junior clerk does in theatre is **not surgery**. It is being a **safe pair of hands**: scrubbing properly, keeping the field sterile, holding a retractor steadily, cutting sutures to the right length, and knowing the instruments well enough to anticipate the surgeon. In the OSCE you will be handed an instrument or a suture packet and asked: *What is it? How do you hold it? What is it used for at CS? What else can it be used for?*

In Uganda, **surgical site infection (SSI)** after CS affects roughly 1 in 10 women in several hospital studies, and the organisms are increasingly **multidrug-resistant**. Much of the prevention is in **your hands**: the scrub, the skin prep, the handling of tissue, the choice of suture, and not contaminating the field.

The one thing never to forget: *if in doubt, it is contaminated.* Change the glove, the gown or the instrument. **Never** close the abdomen on an incorrect count.

These skills are used at **every** caesarean section and laparotomy, and the same principles apply to MVA, vaginal delivery, perineal repair, manual removal of the placenta and cannulation.

Technique	When required
Surgical hand antisepsis (scrub or alcohol surgical rub)	Every operation in theatre, before gowning
Closed gloving	Whenever a gown is worn (theatre)
Open gloving	Procedures without a gown (e.g. catheterisation, VE, perineal repair)
Double gloving	Recommended for all CS in high-HIV/HBV-prevalence settings; mandatory in some units for known HIV/HBV-positive patients
Chlorhexidine-alcohol skin prep	First choice for abdominal skin (WHO)
Povidone-iodine prep	If chlorhexidine is unavailable or she is allergic; the **vagina** before CS in labour
Counts and the WHO Surgical Safety Checklist	Every operation

2. Contraindications and Precautions

Technique	Cautions
Surgical hand antisepsis	Skin broken or infected, so you should not scrub (report it). Sensitivity to chlorhexidine: use povidone-iodine or alcohol rub
Closed gloving	Requires a gown; open gloving otherwise
Double gloving	Slightly reduced dexterity
Chlorhexidine-alcohol skin prep	**Not on mucous membranes** (vagina: use aqueous povidone-iodine). Allow to dry before diathermy. Rare allergy (anaphylaxis)
Povidone-iodine prep	Iodine allergy; must dry to work (about 2 min)
Shaving	Do not shave: it causes micro-abrasions and increases SSI; clip just before surgery if needed
Diathermy	Plate on dry, hairless skin away from metal; never on wet or pooled alcohol prep (fire and burns)

Do not scrub if you have open or infected skin lesions on the hands or a respiratory infection. **Latex allergy** in the patient needs latex-free gloves and catheter for the whole team.

3. Relevant Anatomy and Normal Values

Definitions

Term	Meaning
Asepsis	Preventing microorganisms from entering the surgical wound (sterile instruments, scrub, gowns, gloves, drapes, behaviour)
Antisepsis	Reducing microorganisms on **living tissue** (skin, hands, vagina) with an antiseptic, e.g. chlorhexidine, povidone-iodine, alcohol
Sterilisation	Killing **all** microorganisms including **spores** (autoclave, dry heat, chemical sterilants)
Sterile field	The draped area and the sterile surfaces of the instrument trolley and the scrubbed team, which can be touched only by sterile items and people
Scrubbed team	Surgeon, assistant and scrub nurse (instrument nurse): gowned and gloved
Circulating ("runner") nurse	Unscrubbed. Opens sterile packs, fetches items, ties gowns, and keeps the count record
Suture	Any strand of material used to ligate vessels or approximate tissue
Ligature (tie)	A suture tied around a vessel or pedicle to occlude it
Tensile strength	The force a suture can withstand before breaking. It falls over time as absorbable sutures are broken down
Memory	The tendency of a suture to return to its packaged shape. High memory (nylon, Prolene) means knots slip more easily
Capillarity	Wicking of fluid and bacteria along the strand. It is high in braided sutures

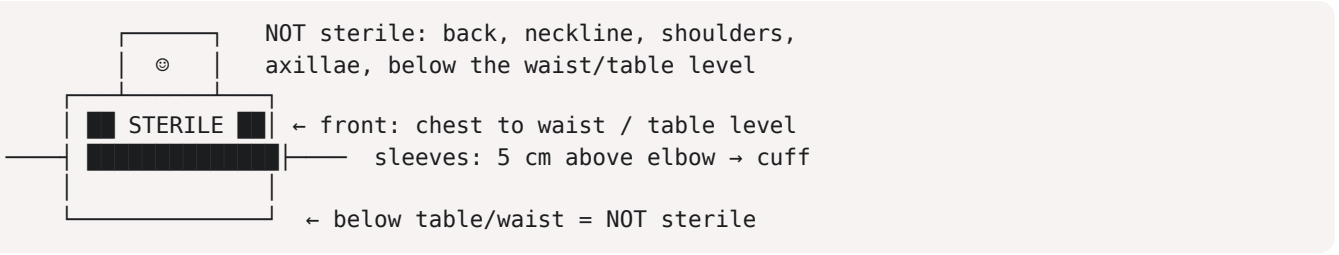
Principles: why asepsis and suture choice matter

- **Where SSI comes from:** mostly the patient's **own skin flora** (*Staph. aureus*, coagulase-negative staph) and, in labour or with ruptured membranes, **vaginal flora** (*E. coli*, *Klebsiella*, anaerobes, GBS). The theatre team adds more through hands, hair, droplets and breaches of the field. **Every breach increases the bacterial inoculum**, and a wound can tolerate only a small inoculum before it becomes infected. Foreign material (sutures, especially braided or non-absorbable) and dead tissue (crushed by clamps, cooked by diathermy) lower the number of bacteria needed.
- **Why the scrub works:** handwashing with antiseptic removes transient flora and reduces resident flora. Chlorhexidine binds to the skin and has **residual (sustained) activity**, which matters because **gloves perforate** during surgery, often unnoticed (in up to a third of long operations) Verify *source for the glove perforation rate*. The scrub is your second line of defence when the glove fails.
- **Why tissue handling matters:** crushing tissue with toothed forceps or clamps, strangulating tissue with tight sutures, and leaving large dead spaces all produce ischaemic tissue where bacteria thrive. Use **atraumatic** instruments on delicate tissues and approximate wound edges without strangling them.
- **Why suture choice matters:** the ideal suture is strong enough for **as long as the tissue needs to heal**, then disappears, provoking minimal inflammation and not harbouring bacteria. The uterus heals quickly (it needs support for a few weeks). The rectus sheath (fascia) heals slowly, regaining only about 50% of strength by 6 weeks, so it needs a suture that holds longer. Skin heals in days.

Abdominal wall layers at a Pfannenstiel CS (outside in)

Skin → subcutaneous fat → **rectus sheath** (opened transversely, held with Kocher's) → rectus muscles (separated) → parietal peritoneum (opened with Metzenbaum) → uterovesical fold and **bladder** (protected by the Doyen) → **lower uterine segment**. Each layer has its own instrument and its own suture (section 4).

The sterile zone of a gowned person



- Keep hands **in front, above the waist, below the shoulders**. **Never fold your arms into your axillae** (the axillae are not sterile) and never drop hands to your sides.
- Scrubbed people pass each other **front-to-front or back-to-back**.
- Unscrubbed people keep **at least 30 cm** from the sterile field, never reach over it, and never walk between two sterile fields.

Key numbers

Item	Value
Surgical scrub	**2–5 min** with antiseptic soap
Scrub area	Fingers to about **5 cm above the elbow**
Skin prep drying time	About **2–3 min**
Antibiotic prophylaxis	Within **60 min before incision**
Fascia strength	Only about **50% by 6 weeks**
Skin suture removal	**Day 5** Pfannenstiel; **day 7–10** midline

4. Equipment and Preparation

The instruments of a CS set

Instrument	How to recognise it	Main use at CS	Other uses / notes
Scalpel handle No. 3	Narrow slot; takes small blades **10, 11, 15**	Skin (No. 10), uterine incision	No. 11 (pointed) for stab incisions (drains); No. 15 for fine work (episiotomy, perineum)
Scalpel handle No. 4	Broader slot; takes large blades **20–24**	Skin and uterine incision (No. 22 is common)	Always load and remove blades with a **needle holder**, never with fingers
Toothed dissecting forceps	Interlocking teeth at the tip (1×2)	Holding **skin, fat, sheath**	Grips firmly without slipping; do not use on bowel or bladder
Non-toothed (plain) dissecting forceps	Serrated tips, no teeth	**Peritoneum, bladder flap, uterus edges, bowel**	DeBakey forceps are an atraumatic version
Mayo scissors, curved	Heavy, short, blunt, curved blades	Cutting the **rectus sheath** and tough tissue	Curved blades allow dissection along planes

Instrument	How to recognise it	Main use at CS	Other uses / notes
Mayo scissors, straight	Heavy, straight	**Cutting sutures** ("suture scissors")	Also cutting dressings and drapes
Metzenbaum scissors	Long, slender handles, short fine blades	**Delicate dissection**: opening peritoneum, uterovesical fold, bladder flap	**Never cut sutures** with them (it blunts them)
Artery forceps (Spencer Wells / haemostats), straight and curved	Ratchet, fine transverse serrations on the full jaw	Clamping bleeding vessels in fat and sheath; **lifting the peritoneum** to open it; holding sutures	Mosquito forceps are a smaller, finer version
Kocher's forceps	Heavy straight artery forceps with **teeth at the tip** (1×2)	Gripping and lifting the **rectus sheath** edges (Pfannenstiel)	Heavy fibrous tissue; also used to grasp the cervix in some procedures
Allis tissue forceps	Ratchet; slightly curved jaws ending in **fine interlocking teeth**	Holding **sheath edges** and fascia; holding the edges of the uterine incision if Green-Armytage are unavailable	Traumatic: only for tissue that is tough or will be excised. Also used for the cervix (VVF repair) and in hysterectomy
Babcock forceps	Ratchet; **rounded fenestrated (windowed) atraumatic tips**, no teeth	Lifting the **fallopian tube** for tubal ligation (Pomeroy)	Holding bowel, ureter, appendix, round ligament: **tubular structures**
Green-Armytage forceps	Large, long flat haemostatic clamps with broad serrated jaws	Clamping the **angles and bleeding edges of the uterine incision** and holding them for closure	Specifically obstetric; reduce blood loss from the lower-segment edges
Littlewood's tissue forceps	Wide jaws with a few (three) teeth; lightweight	Holding **skin or peritoneal edges**	Sometimes in the set instead of Allis
Sponge-holding forceps (Rampley / Foerster ring forceps)	Long, ring-shaped serrated jaws	**Skin prep** (holding the swab); swab on a stick for mopping	Holding the cervix (evacuation, PPH exploration); removing products of conception
Towel clips (Backhaus)	Small ratchet with sharp curved points	**Securing drapes**	Can pierce the skin, so place them on drapes
Needle holder (Mayo-Hegar)	Short, stout jaws with **cross-hatched grooves**, often a gold handle (tungsten carbide)	Holding the curved needle for all suturing	Hold the needle **one-third to half-way from its swaged end** at the tip of the jaws
Doyen retractor	**Single curved, broad blade** on a handle	Placed over the pubis to **retract and protect the bladder** and expose the lower segment	Removed before delivering the head
Langenbeck retractors	Small hand-held right-angled blades	Retracting **skin, fat and sheath** edges	Superficial retraction; often used in pairs
Morris retractor	Larger, broader right-angled blade	Retracting the **abdominal wall** (rectus) laterally	Deeper than Langenbeck
Deaver retractor	Large, broad, curved "question-mark" blade	Deep retraction of the abdominal wall or bowel	More often in the hysterectomy or laparotomy set
Self-retaining retractor (Balfour / Gosset)	Frame with two lateral blades ± a bladder blade; ratchet	Holding the abdominal wall open at hysterectomy or a difficult CS	Frees the assistant's hands

Instrument	How to recognise it	Main use at CS	Other uses / notes
Suction: Yankauer	Rigid curved tube with a bulbous tip	Superficial suction; clearing the **baby's mouth** (gently) or maternal airway	–
Suction: Poole	Straight tube with a **perforated outer sheath**	Sucking liquor and blood from the **peritoneal cavity**	The guard stops omentum blocking it
Wrigley's forceps (sometimes)	Short obstetric forceps	Delivering a **high floating head** at CS	Outlet forceps delivery
Swabs / abdominal packs	Gauze with a **radio-opaque (X-ray detectable) marker** and a tape	Mopping, packing bowel away	**Counted**; tape ends stay outside the wound
Kidney dish, gallipots, bowls	–	Neutral zone for sharps; antiseptic; saline	Pass the **scalpel in a kidney dish** (hands-free)
Cord clamps / artery forceps and cord scissors	–	Clamping and cutting the cord	–
Diathermy pencil and plate	–	Haemostasis in fat, sheath	The plate must be on dry skin away from metal; beware alcohol prep (burns and fire)

EXAM FAVOURITE

For an **instrument station**, answer in this order: **name, class, identifying features, how you hold it, use at CS, one other use, and a precaution**. For example: "This is a **Babcock forceps**, a tissue-holding forceps with a ratchet and **atraumatic fenestrated tips**. I hold it with thumb and ring finger in the rings and the index finger on the shaft. At CS it is used to **lift the fallopian tube for tubal ligation**. It is also used to hold bowel. Because it is atraumatic it is suitable for delicate tubular structures."

How to hold and pass instruments

- **Ring-handled instruments** (scissors, needle holder, artery forceps): **thumb and ring finger** in the rings, the **index finger** along the shaft for control (the "tripod grip"). Don't push fingers through to the knuckle, or you can't release quickly.
- **Dissecting forceps**: held like a **pen** between thumb and index/middle finger. Never in a fist.
- **Scalpel**: a **"table-knife" grip** for long skin incisions, and a **pen grip** for fine, controlled cuts (the uterine incision).
- **Passing**: the scrub nurse slaps the handle **firmly into the surgeon's palm**, curved tips pointing the way they will be used, ratchet closed. **Sharps are passed in a kidney dish** (the "neutral zone"), never hand to hand. This is one of the most effective ways to reduce needle-stick injuries.
- **Retracting (the assistant's main job)**: hold steady, don't lean, don't let go without telling the surgeon, and follow the surgeon's focus. Pull **gently**: over-retraction tears the rectus or bruises the bladder.

Sutures: classification and properties

Suture (brand example)	Type	Structure	Tensile strength retention	Absorption complete	Typical use in O&G
Plain catgut	Absorbable, **natural** (sheep/cattle intestine collagen)	Twisted "monofilament-like"	About **7–10 days**	About **70 days** (enzymatic, unpredictable; faster with infection)	Tubal ligation (Pomeroy), subcutaneous ties, peritoneum
Chromic catgut	Absorbable, natural, treated with chromium salts (slower)	Twisted	About **2–3 weeks**	About **90 days**	**Uterus (#1 or #2)** in many Ugandan theatres, perineum (older practice)
Polyglactin 910 (Vicryl)	Absorbable, **synthetic**	**Braided**, coated	About **75% at 2 weeks, 50% at 3 weeks**	**56–70 days** (hydrolysis, predictable)	**Uterus (#1), rectus sheath (#1)**, bladder repair (2-0/3-0), subcutaneous (2-0), subcuticular (3-0)
Vicryl Rapide	Absorbable, synthetic	Braided	About **50% at 5 days**, gone by 10–14 days	About **42 days**	**Perineal repair / episiotomy** (less pain, fewer suture removals)
Poliglecaprone 25 (Monocryl)	Absorbable, synthetic	**Monofilament**	About **50–60% at 1 week**, 20–30% at 2 weeks	**91–119 days**	**Subcuticular skin closure** (3-0/4-0)
Polydioxanone (PDS)	Absorbable, synthetic	**Monofilament**	Useful strength for about **6 weeks** (long)	About **6 months** (182–238 days)	**Rectus sheath (#1 loop)**, especially in high-risk wounds (obesity, infection, midline)
Nylon (Ethilon)	**Non-absorbable**, synthetic	**Monofilament**	Long-lasting (degrades very slowly)	Permanent (removed)	**Skin**, interrupted or mattress (2-0/3-0); loop nylon for mass closure
Polypropylene (Prolene)	Non-absorbable, synthetic	Monofilament	Permanent	Permanent	Mass closure, vascular repair, pull-out subcuticular
Silk	Non-absorbable, **natural**	**Braided**	Loses strength over about a year	Encapsulated	Ties, securing drains and catheters; rarely skin now (infection, reaction)

CLINICAL PEARL

Monofilament vs braided: the trade-off. Braided sutures (Vicryl, silk) are **soft, easy to handle and knot securely**, but their interstices **wick fluid and shelter bacteria** (capillarity). They cause more infection when used in a contaminated wound, and non-absorbable braided silk can cause **stitch sinuses** for months. Monofilaments (nylon, PDS, Monocryl, Prolene) glide through tissue with **less drag and less infection**, but have **memory**. Their knots slip, so they need **more throws** (about 5–6 for nylon versus 3–4 for braided).

Gauge and needles

USP gauge: the more zeros, the thinner. From thickest to thinnest: **2 > 1 > 0 > 2-0 > 3-0 > 4-0 > 5-0 …**

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Thick ──────────────────────────────────────────────► Thin
  #2     #1      0      2-0      3-0      4-0     5-0
 (rare) uterus  uterus  subcut/  skin     fine    vessels/
        sheath  sheath  peritoneum subcut. skin   paeds
```

Needle	Point / cross-section	Use	Why
Round-bodied (taper point)	Round, tapers to a sharp point, no cutting edge	**Uterus, peritoneum, bowel, bladder, subcutaneous tissue**	**Spreads** tissue fibres rather than cutting, so it leaves a small hole that seals, with less bleeding and leakage
Cutting	Triangular; cutting edge on the **inner (concave)** curve	**Skin**	Cuts through tough skin. It may cut out towards the wound edge
Reverse cutting	Triangular; cutting edge on the **outer (convex)** curve	**Skin**, tough fascia	Less likely to cut out through the wound edge. The most common skin needle today
Taper-cut (trocar point)	Cutting tip, round body	Tough fibrous tissue (sheath, cervix, scarred tissue)	Penetrates easily, then spreads
Blunt-point (taper)	Rounded blunt tip	Friable tissue (liver); **rectus sheath closure** in some units	**Reduces glove puncture and needle-stick**

Needle shape: **½ circle** is the workhorse for deep tissue (uterus, sheath). **⅜ circle** is used for skin. **Straight (hand-held) needles** are unsafe (needle-stick) and outdated. Needle size is described by its chord length (e.g. 40 mm for uterus, 24 mm for skin).

Which suture for which layer at CS (typical Ugandan practice)

Layer	Suture	Needle	Technique	Why
Uterus (2 layers)	**#1 Vicryl** or **#1/#2 chromic catgut**	Round-bodied, ½ circle, about 40 mm	1st layer **continuous locking** (haemostatic); 2nd layer continuous **imbricating** (burying the first)	Strength for the weeks of healing. Absorbable because it lies inside the peritoneum. A 2-layer closure may reduce later rupture
Visceral / parietal peritoneum (if closed)	2-0 plain or chromic catgut, or 2-0 Vicryl	Round-bodied	Continuous	NICE says non-closure is acceptable
Rectus sheath	**#1 Vicryl** or **#1 PDS (loop)**; loop nylon in some units	Round-bodied / taper-cut / blunt	**Continuous**, bites about 1 cm from the edge, 1 cm apart; not too tight	Fascia regains strength slowly, so it needs a long-lasting suture. Continuous is fast and distributes tension
Subcutaneous fat (only if more than 2 cm)	2-0 plain catgut or 2-0/3-0 Vicryl	Round-bodied	Interrupted	Closing thick fat reduces seroma and separation
Skin	**3-0 Monocryl/Vicryl subcuticular**, OR **2-0/3-0 nylon interrupted or mattress**, OR staples	Reverse cutting, ⅜ circle	Subcuticular, or interrupted / vertical mattress	Subcuticular is cosmetic and avoids removal. Nylon is used where infection risk is high (it can be removed to drain)
Tubal ligation (Pomeroy)	**#0 or #1 plain catgut**	Round-bodied	Tie the base of a loop of tube, excise the loop	Plain catgut absorbs **quickly** so the cut ends separate and do not recanalise
Bladder repair	2-0/3-0 Vicryl (or chromic)	Round-bodied	**Two layers**, the first avoiding the mucosa	Absorbable (non-absorbable sutures in the bladder form **stones**). Catheter for 7–14 days

UGANDA CONTEXT

Chromic catgut is **banned in Europe** (BSE concerns) but is still widely used in Ugandan public hospitals because it is cheap. Vicryl is often supplied by the patient ("buy these sutures from the pharmacy" lists). Know both. When catgut is used on the uterus, remember that its absorption is **faster and less predictable in infected tissue**. That is one reason synthetic sutures are preferred when chorioamnionitis is present, if they are available.

Preparation: the first 5 minutes when called to an emergency CS

1. **Find out the category and the indication** ("Category 1 for cord prolapse"). In a Category 1 case, work fast but **do not skip the scrub, the prep drying time or the count**. These take minutes and prevent disasters.
2. **Check yourself**: no open cuts or skin infection on your hands, no respiratory infection; cap, mask, **eye protection**, apron, theatre shoes; jewellery off.
3. **Look at the patient on the table**: is she awake and talking (spinal) or being intubated (GA)? Pulse and BP on the monitor, **15° left tilt**, IV lines running, **catheter draining**, diathermy plate on.
4. **Join the WHO Sign in** (before anaesthesia): identity, procedure, **consent**, allergies, airway or aspiration risk, **expected blood loss more than 500 mL** with two IV lines and blood available.
5. **Open your gown and glove packs** (or ask the circulator) and go to scrub.

Focused history: what to ask before you scrub, and why

Ask these from the notes, the patient and the anaesthetist. Each answer changes something you do.

Question	Why
Allergies (latex, iodine, chlorhexidine, penicillin)	Glove choice, skin-prep choice and antibiotic choice. A latex allergy needs latex-free gloves and catheter for the whole team
HIV and hepatitis B status (if known)	Double gloving, extra sharps discipline, eye protection. Standard precautions apply to **every** patient anyway
Labour, ruptured membranes, number of VEs, fever	A contaminated field needs **vaginal povidone-iodine cleansing** and possibly **therapeutic** rather than prophylactic antibiotics; favours synthetic monofilament sutures for the sheath
Was the antibiotic given, and when?	It must be given **within 60 minutes before the skin incision**; confirmed at the Time out
BMI, diabetes, steroids, anaemia, HIV with a low CD4	Poor wound healing: consider **PDS for the sheath**, **close fat more than 2 cm**, careful skin closure, check glucose
Previous abdominal surgery and scars	Adhesions: the surgeon may need Metzenbaum scissors, extra retractors or a midline incision
Was the skin shaved on the ward?	Shaving causes micro-abrasions and more SSI. Tell the surgeon, and teach the ward to **clip instead**
Any sutures the family bought (common in Ugandan public hospitals)	Check what is actually available (#1 Vicryl? chromic?) **before** the incision, not at closure
Your own health	Open skin lesions or a respiratory infection mean don't scrub

Examination of the patient, the equipment and yourself

The patient's skin site (look before the prep): hair (clip only if it gets in the way), **dirty umbilicus** (clean first), rash, skin infection or intertrigo in the skin fold, shaving abrasions, old scars (type and site).

The equipment (this is the "examination" a scrub assistant must be good at):

Item	What to check	If abnormal
Instrument and drape packs	**Indicator tape changed colour**, pack **dry, intact, in date**, labelled with the date and "sterilised by"	Wet, torn, dropped or unchanged tape: **not sterile**. Replace it
Glove and gown packs	Intact, dry, in date, correct sizes	Replace
The CS set on the trolley	All expected instruments present: scalpel handles and blades, toothed and non-toothed dissecting forceps, Mayo (curved and straight) and Metzenbaum scissors, artery forceps, **Kocher's**, **Allis**, **Babcock**, **Green-Armytage**, sponge-holders, towel clips, **needle holder**, **Doyen**, Langenbeck, suction	Ask for missing items **before** the incision
Sutures	Read each packet: **material, gauge, needle type, length, expiry**. #1 round-bodied for uterus and sheath; 3-0 on a cutting needle for skin	Missing #1 suture = the operation cannot be closed safely
Suction	Working, tubing connected, Yankauer and Poole available	Fix before incision; you will need it for liquor and blood
Diathermy	Plate on **dry, hairless skin** (thigh) away from metal; pencil working	Poor plate contact causes burns
Swabs	**Radio-opaque**, counted, with tapes on packs	Never use non-radio-opaque gauze inside the abdomen

Yourself: gloves for holes (check them against the light when you change them), sleeves and cuffs covered, no hair outside the cap.

Bedside checks before and during the case

- **Chemical indicator tape** on every pack and, where used, an **internal indicator strip** inside the pack: has it changed colour?
- **Glove integrity**: after any sharp contact, look for holes; with **double gloving**, a coloured inner glove makes a perforation visible.
- **Skin-prep dry check**: alcohol prep **dry to touch and no pooling** under the patient before diathermy (about 2–3 minutes).
- **Baseline count** of swabs, needles, blades and instruments, written on the board.
- **HemoCue Hb** and **random blood sugar** (poor glucose control and anaemia increase SSI).
- **Pregnant or other patients**: urine dipstick, malaria RDT if febrile.

5. Procedure Step by Step

A. Before scrubbing

1. Theatre clothing on: **scrub suit, cap covering all hair, mask covering nose and mouth** (tie the top strings over the ears and the bottom at the neck), **eye protection**, theatre shoes or boots, and a **plastic apron** (for CS, because of liquor and blood).
2. **Remove all jewellery** (rings, watch, bracelets). Nails should be **short, with no polish or artificial nails**. These harbour bacteria and puncture gloves.

3. Check your hands for **cuts or skin lesions**. Cover minor cuts with a waterproof dressing and report open or infected lesions: you should not scrub.
4. Open your **gown and glove packs** onto a separate clean surface (or ask the circulator to) with correct no-touch technique. Check the **sterility indicator, expiry and integrity** of the packs.

B. Surgical hand antisepsis (WHO 2009)

Option 1: Antiseptic soap and water ("the scrub")

1. Adjust the water with the elbow or foot tap. Wet hands and forearms.
2. **First scrub of the day**: clean **under the nails** with a nail cleaner under running water. **Brushes are not recommended** on the skin: they damage it and increase shedding. A soft sponge or brush may be used for nails only.
3. Apply antiseptic soap (**chlorhexidine gluconate 4%** or **povidone-iodine 7.5%**). Wash systematically: **each finger has 4 sides**, then the web spaces, palms and backs, wrists, and forearms up to about **5 cm above the elbow**.
4. **Total contact time about 2–5 minutes** (follow the product's instructions). Longer scrubs (10 min) give no extra benefit and damage the skin.
5. Keep your **hands above the elbows** at all times, so water runs from the clean fingertips down to the less clean elbows.
6. **Rinse from fingertips to elbow** in one direction, passing each arm through the water once. Do not shake the hands.
7. Walk into theatre **backwards through the door**, hands held up in front of you (above waist, below shoulders), away from the body.
8. **Dry with a sterile towel**: one end for one hand and forearm (fingers to elbow, blotting and rotating, never going back), then the other end for the other arm. Drop the towel without touching anything.

Option 2: Alcohol-based surgical hand rub (WHO-preferred where water quality is doubtful)

1. On arrival, **wash hands with plain soap and water** and dry them (to remove spores such as *Clostridioides*).
2. Apply the rub (about 5 mL at a time) to the **fingertips of one hand** (dip them in the palm), then spread it over the forearm up to the elbow with a circular motion, and repeat for the other hand.
3. Then rub the hands with the 6-step hygienic technique (palm to palm, backs, interlaced, backs of fingers, thumbs, fingertips) until dry.
4. Keep the skin **wet for the full contact time** in the manufacturer's instructions (typically **about 1.5–3 minutes**). **Let it dry completely** before gloving.

C. Gowning

1. Pick up the folded gown **by the inside, at the neckline**, and lift it clear of the trolley.
2. Step back into an open space and let it **unfold, inside facing you**. Do not shake it.
3. Slip both arms into the sleeves **without your hands coming out through the cuffs** (for closed gloving).
4. The **circulating nurse** pulls the gown on from behind, touching only the inside, and ties the **neck and inner waist ties**.

D. Closed gloving

1. With hands still inside the sleeves, pick up the **first glove** through the sleeve. Place it **on the palm-up forearm**, with the **glove's palm against your palm** and its **fingers pointing towards your elbow**.
2. Through the sleeve, grasp the **bottom rim** of the glove with the thumb of that hand. With the other (covered) hand, grasp the **upper rim** and stretch it **over the back of the hand and the cuff**.
3. **Push the hand through** into the glove while pulling the glove and sleeve up together. Adjust the fingers.

4. Repeat for the second hand using the gloved hand. **The knitted cuff must be completely covered.**
5. **Double gloving**: put a second pair on over the first (open technique is acceptable over the first gloved pair). A coloured inner "indicator" glove shows perforations.
6. **Tie the wrap-around (outer) tie**: hand the card or tag to the circulator (or a gloved colleague), turn 360° so the flap wraps your back, take back the tie, and tie it at your side.

Open gloving (no gown, e.g. VE, catheterisation): pick up the first glove by its **folded cuff (inside)** only, then slide the gloved fingers **under the cuff (outside)** of the second glove.

E. Skin preparation

1. **Hair**: leave it, or **clip** it just before surgery. **Never shave** (shaving increases SSI).
2. **Clean the umbilicus** first if it is dirty (it is a reservoir).
3. For CS in labour or with ruptured membranes, **vaginal cleansing with povidone-iodine** (by the surgeon, before the abdominal prep) reduces post-op endometritis.
4. The scrubbed surgeon or assistant uses **sponge-holding forceps** and gauze soaked in **2% chlorhexidine gluconate in 70% alcohol** (WHO first choice) or **10% povidone-iodine**.
5. Paint from the **incision site outwards**. Use **back-and-forth friction for about 30 seconds** for chlorhexidine, or **concentric circles from centre to periphery** for iodine. **Never return** from the periphery to the centre with the same swab.
6. **Area**: from the **xiphisternum to mid-thigh**, and laterally to the **flanks (mid-axillary lines)**, including the pubis. Wide enough to extend the incision or convert it to a midline one.
7. Use **2 or 3 fresh swabs**, and discard each forceps and swab once it has touched the periphery.
8. **Let it dry fully (about 2–3 minutes)**. Alcohol is **flammable**: diathermy on wet alcohol or pooled prep under the patient causes **fires and burns**.

F. Draping

1. Drapes are handled **high and away from you**, with your gloved hands protected inside a **cuff** of the drape.
2. Place **four towel drapes** to "square off" the incision site (near side first, or as per unit practice), secured with **towel clips**. Then place the large **laparotomy (fenestrated) drape** with the opening over the incision, unfolding it towards the head and then the feet.
3. **A drape, once placed, is not moved towards the incision.** It can only be moved outwards (moving it inwards drags skin bacteria in). If it is misplaced, **discard it** and use a new one.
4. The part of a drape that falls **below the table edge** is non-sterile. Don't lift it back up.
5. **Wet drapes are contaminated** (strike-through). Cover with an extra drape.

G. Maintaining the sterile field during the operation

- **Sterile touches sterile only.** If in doubt, it is **not** sterile.
- **Anything that falls below the table or onto the floor is contaminated.** Leave it and ask for another.
- Keep **doors closed** and traffic and talking to a minimum (each person sheds thousands of skin particles per minute).
- **Glove torn or punctured**: step back, have the glove changed (the scrub nurse pulls it off and regloves you by assisted gloving), and if it was a sharps injury, follow the needle-stick protocol after the case.
- **Gown sleeve contaminated**: cover it with a sterile sleeve or change the gown.
- Use a **neutral zone** (kidney dish) for sharps. **Never recap needles by hand.** Announce "sharp" when passing.
- **Separate "dirty" instruments** used on the vagina or cervix (e.g. for disimpacting the head) from abdominal instruments. **Change gloves** after putting a hand into the vagina.
- **Anticipate as the assistant**: after the sheath is opened, the surgeon will want **Kocher's**. Before the uterine incision, a **Doyen** and a **fresh blade**. After delivery, **Green-Armytage** clamps and a **#1 round-bodied suture**.

Before closing, **the count**.

- **Cut suture tails** to **3–5 mm** for braided and **5–10 mm** for monofilament, with **straight Mayo** scissors.

H. Counts and the WHO Surgical Safety Checklist

When	What
Before incision (baseline)	Scrub nurse and circulator count **swabs, packs, needles, blades and instruments aloud together** and write them on the count board
Items added during surgery	Counted in and recorded immediately
Before closing the uterus (a cavity)	First closing count
Before closing the peritoneum or rectus sheath	Second closing count
At skin closure / end	Final count. The surgeon is told **"count correct"**

WHO Surgical Safety Checklist (3 pauses):

1. **Sign in** (before anaesthesia): identity, procedure, consent, site; anaesthesia machine checked; allergy; airway or aspiration risk; blood loss risk (more than 500 mL) with IV access and blood ready.
2. **Time out** (before skin incision): team introductions; confirm patient, procedure and **CS category**; **antibiotic given within the last 60 min?**; anticipated critical events; sterility confirmed; essential imaging.
3. **Sign out** (before the patient leaves theatre): procedure recorded; **counts correct**; specimens labelled; equipment problems; key concerns for recovery.

I. After the operation

1. **Sharps into the safety box** by the user (the person who used it disposes of it).
2. Instruments go into **0.5% chlorine solution for 10 minutes** (decontamination), then are cleaned, packed and **autoclaved** (121 °C at 106 kPa for 20 min unwrapped or 30 min wrapped). See the IPC topic.
3. Remove PPE in the correct order (gown and gloves together, rolled inside out; then eye protection; then mask), and perform hand hygiene.

J. Post-operative wound review (Hutchison order)

1. **Introduce yourself, consent, hand hygiene**; position supine and expose the wound only.
2. **General inspection and vitals first**: pulse, BP, temperature. A **pulse more than 100, fever or low BP** means you may be dealing with sepsis or bleeding, not just a "wound issue".
3. **Look at the dressing**: **fresh blood** (haematoma or bleeding), **pus** (SSI) or **pink serous fluid** on day 5–10 (**warning sign of fascial dehiscence**, a burst abdomen).
4. **Inspect the wound**: edges, redness (extent in cm), swelling, discharge, gaping.
5. **Palpate**: warmth, induration, fluctuation, crepitus, tenderness.
6. **Probe the wound gently** with a sterile cotton-tipped swab after removing 1–2 sutures if there is pus: does it drain, is the **sheath intact**?
7. **Look for other sources of fever**: uterus, urine, breasts, chest, calves, malaria.
8. Re-dress aseptically, thank her, hand hygiene, document.

Questions to ask at the wound review: wound pain getting worse after day 2 (normal pain gets better); discharge and its colour; fever or rigors; when the dressing was last changed; diabetes or anaemia; whether she is eating and mobilising.

6. Findings and Interpretation

Post-op wound findings (day 2–7)

Finding	Interpretation
Clean, dry, edges together, mild tenderness	Normal healing
Redness **under 1–2 cm** around sutures only	Suture reaction: watch
Spreading redness, warmth, induration, increasing pain	Cellulitis or early SSI
Pus from the wound or a suture	**Superficial SSI** or stitch abscess
Fluctuant swelling under the wound	Haematoma or abscess: may need opening
Pink ("salmon") serous discharge, day 5–10	**Warning of fascial dehiscence (burst abdomen)**
Gaping wound with bowel or omentum visible	**Burst abdomen: emergency**
Crepitus, dusky skin, severe pain, toxic patient	**Necrotising fasciitis: emergency surgical debridement**

Laboratory and imaging

Test	When	Expected / abnormal finding	How it changes management
Hb / FBC (HemoCue at HC IV; FBC at hospital)	Before surgery; any wound problem	Anaemia (Hb under 11 g/dL; severe under 7); raised WBC with infection	Anaemia delays healing: iron or transfusion; WBC supports infection
Random blood sugar (all levels)	Diabetes, obesity, steroids, poor healing	High glucose	Glucose control reduces SSI (WHO SSI guideline)
HIV and HBsAg (rapid tests at HC III/IV and above)	Unknown status	Positive	Double gloving; ART; also informs the needle-stick plan if a sharps injury happens
Pus swab for culture and sensitivity (microbiology at RRH/national referral; rarely at HC IV)	Any discharging wound, **taken before antibiotics**	Often **ESBL *Klebsiella*/*E. coli*** (ceftriaxone-resistant) or **MRSA**	Targeted antibiotics; IPC alert if several cases
Blood cultures (RRH/national)	Fever with systemic signs	Bacteraemia	IV antibiotics per sensitivity
Ultrasound of the wound and pelvis (many hospitals)	Fluctuant swelling, persistent fever	Subcutaneous or **subsheath collection**, pelvic abscess	Drainage (bedside or in theatre)
Plain abdominal X-ray (hospitals with radiology; sometimes a C-arm in theatre)	**Incorrect count** after a full search; late mass, sinus or fever after surgery	Radio-opaque marker of a retained swab	Re-explore; prevents a gossypiboma
Biological indicator (spore test) of the autoclave (national/regional sterilisation units, where available)	Routine periodic check; any cluster of SSIs	Growth = sterilisation failure	Stop using the autoclave, recall packs, repair
Theatre and autoclave logs (every facility)	SSI cluster	Missing cycles, short times, unchanged indicators	Fix the process; retrain staff

UGANDA CONTEXT

Culture and sensitivity is not available at most HC IVs, and results take 3–5 days even at referral hospitals. So **take the swab before starting antibiotics**, start a sensible empirical regimen for a spreading or systemic infection, and **change to the culture result**. Local SSI isolates are often ceftriaxone-resistant, so "more ceftriaxone" is often the wrong answer.

Red flags

RED FLAG

- **Count incorrect**: STOP closure. Search the wound, the field, the floor and the bins. Take an X-ray if available (radio-opaque markers). Never close "and hope".
- **Alcohol prep still wet or pooled** when diathermy is about to be used: fire risk. Wait and blot.
- **Sharps injury** to anyone in theatre: finish safely, then report, first aid and PEP assessment **within hours** (not days).
- **Pack sterility indicator not changed / pack wet or torn**: the pack is **not sterile**. Replace it.
- **Unexplained post-op fever with a sinus or chronic wound discharge weeks later**: think of a **stitch sinus (braided non-absorbable) or a retained swab (gossypiboma)**.

7. Clinical Acumen

The "acumen" in theatre skills is **judgement**: noticing breaches, choosing materials sensibly, and anticipating.

Thinking like a clinician

1. **Choosing a suture: ask 4 questions.** (a) How **long** does this tissue need support? (fascia: long; uterus: medium; skin: short). (b) Is the field **contaminated**? (then prefer monofilament and avoid braided non-absorbable). (c) Will it be **removed**? (skin: nylon or staples; inside: absorbable). (d) What is **available and affordable** for this patient?
2. **Choosing a needle**: soft tissue that parts easily gets **round-bodied**. Skin or tough tissue gets **(reverse) cutting**.
3. **Choosing forceps**: will this tissue be **kept or removed**? Tissue that stays (bowel, tube, bladder) gets **atraumatic** forceps (Babcock, non-toothed). Tough tissue or tissue to be excised gets **toothed** forceps (Allis, Kocher, toothed dissectors).
4. **Anticipating as an assistant**: after the sheath is opened, the surgeon will want **Kocher's**. Before the uterine incision, a **Doyen** and a **fresh blade**. After delivery, **Green-Armytage** clamps and a **#1 round-bodied suture**. Before closing, **the count**.
5. **A "wound problem" is a patient problem first**: vitals before the dressing.
6. **Honesty about a breach** is a safety behaviour, not an embarrassment: say it, fix it, document it.

Common student mistakes

- Scrubbing with rings on.
- Touching the mask or cap after scrubbing.
- Letting hands drop below the waist; folding arms under the axillae.
- Touching the outside of the gown while donning it.
- Reaching across the field.
- Cutting sutures with Metzenbaum.

- Cutting sutures too short (knots unravel) or too long (irritation). Cut **3–5 mm** tails for braided and **5–10 mm** for monofilament.

Worked examples

Worked example 1: an incorrect count. At sheath closure the scrub nurse says, "We have 19 swabs; the board says 20." *Interpretation:* one swab is unaccounted for, and it may be in the abdomen. *What to do:* **stop closing**; the nurse **recounts**; the surgeon **explores** the paracolic gutters, the pouch of Douglas, under the uterus and the subhepatic space; the circulator searches the **floor, bins, drapes and linen**. Here the swab is found in a bucket under the table and the count is now 20 = 20. **Document**, then close. If it had not been found: **X-ray before she leaves theatre** (radio-opaque marker) and an incident report.

Worked example 2: reading a wound on day 5. A woman on day 5 after an emergency CS for prolonged labour has a temperature of **38.2 °C**, pulse 104, and a Pfannenstiel wound that is red and indurated for 3 cm on each side, with pus at the right end. Nylon interrupted skin sutures were used. *Interpretation:* a **superficial incisional SSI** with systemic signs. *Plan:* **pus swab for culture**; **remove the sutures over the collection** to drain it; **probe** to check the sheath is intact (if the sheath has opened, it is a deep SSI or dehiscence and she needs theatre); daily dressing; empirical antibiotics because she is febrile and tachycardic (per the unit protocol, reviewed with the culture); check **Hb and RBS**; and look for other causes of fever (uterus, urine, breasts, chest, calves, **malaria**). Three days later the culture grows **ESBL *Klebsiella* resistant to ceftriaxone**: change to an antibiotic it is sensitive to, and tell the **IPC focal person** if other women have similar infections.

8. Complications and Troubleshooting

Here "management" means handling the problems that arise from instruments, sutures and asepsis.

1. Breach of sterility (step by step)

1. **Say so immediately.** Owning a breach protects the patient. Nobody should be embarrassed.
2. **Glove breach**: step away from the field and hold out the hand. The scrub nurse removes the glove and regloves you (assisted gloving). If both gloves are breached, or you were cut, change both and **report a sharps injury**.
3. **Gown breach** (e.g. brushed an unsterile surface): cover the area with a sterile sleeve, or change the gown **and** gloves.
4. **Contaminated instrument**: remove it from the field and use a sterile replacement.
5. **Contaminated field** (e.g. drape soaked through): add a sterile drape over it.
6. **Document** significant breaches, and consider whether extra antibiotic cover is needed (rarely).

2. Incorrect count

1. The surgeon stops closing. The scrub nurse repeats the count.
2. The surgeon **systematically explores** the cavity (gutters, under the uterus, the vesico-uterine pouch, the subhepatic space).
3. The circulator searches **floor, bins, drapes and linen**.
4. If still missing: **intra-operative X-ray** where available (radio-opaque marker), or before she leaves theatre.
5. **Document** and report as an incident. Inform the patient if an item was retained.

> CLINICAL PEARL
> A **retained swab (gossypiboma)** presents weeks to years later with fever, a mass, a discharging sinus, bowel obstruction or a fistula. It is entirely preventable with disciplined counts, **radio-opaque swabs**, and **tapes on packs left outside the wound**.

3. Sharps injury in theatre

- **First aid immediately**: let it bleed freely (don't squeeze), wash with soap and running water, and flush mucous membranes with water or saline.
- Report to the in-charge; test the **source patient** (with consent) and yourself.
- **PEP (TDF/3TC/DTG once daily for 28 days)** started **as soon as possible, ideally within 2 hours and no later than 72 hours**. Hepatitis B vaccination or booster if not immune. Full detail is in the IPC topic.

4. Suture-related wound complications

Problem	Features	Management
Stitch abscess	Localised red, tender pustule at a suture	Remove that suture, drain, dress. Antibiotics only if cellulitis spreads
Wound infection (SSI)	Erythema, discharge, fever on day 3–7	Remove some skin sutures to **drain**, send pus for culture, daily dressing, antibiotics if cellulitis or systemic signs
Wound dehiscence / burst abdomen	Serosanguinous ("pink") discharge on day 5–10, then gaping or bowel visible	Cover with **sterile moist saline packs**, reassure, keep NPO, IV fluids and antibiotics, **return to theatre** for mass closure (e.g. loop nylon or #1 PDS). Risks: infection, anaemia, obesity, poor technique, sutures tied too tight (they cut through)
Stitch sinus	Chronic discharging point weeks to months later, from braided non-absorbable (silk)	Explore and remove the suture
Suture "spitting"	Absorbable subcuticular suture extruding	Trim; it resolves
Hypertrophic or keloid scar	Raised scar (commoner in African skin)	Reassure; intralesional steroid if troublesome; tension-free closure and subcuticular sutures help
Necrotising fasciitis	Crepitus, dusky skin, severe pain, toxic patient	Emergency surgical debridement, IV antibiotics, resuscitation

5. Skin suture removal (timing)

- **Pfannenstiel / transverse**: **day 5** (or day 3–5 for staples).
- **Midline**: **day 7–10** (longer if obese, anaemic or infected). Remove **alternate sutures first**, and remove the rest a day or two later if the wound is holding.
- Technique: clean, lift the knot with forceps, cut the suture **close to the skin on one side** (so no exposed, contaminated suture is dragged through), and pull through towards the wound.

6. Allergic reaction and prep burns

- **Chlorhexidine anaphylaxis** (rare but real): stop exposure and give **adrenaline 0.5 mg IM** (0.5 mL of 1:1000), oxygen, fluids and antihistamine. Use povidone-iodine next time and document it.
- **Chemical or diathermy burns** (pooled prep, poor plate contact): prevent them by drying the prep and placing the plate on dry, hairless, well-vascularised skin (the thigh) away from metal.

7. Equipment troubleshooting

Problem	Fix
Pack indicator not changed, wet, torn or dropped	Not sterile: replace
#1 suture not available before incision	Tell the surgeon now; send for it before the skin is cut

Problem	Fix
Suction not working	Fix before incision (tubing, connection, canister)
Poole blocked by omentum	The perforated guard prevents this; check it is fitted
Drape misplaced towards the incision	Do not drag it in; discard and replace
Drape soaked through	Cover with an extra sterile drape

9. Documentation and Presenting Findings

Model theatre-readiness summary (what you say at the Time out as the assistant): "Mrs N., 28, emergency LSCS Category 2 for obstructed labour. **Ceftriaxone 2 g and metronidazole 500 mg given 20 minutes ago.** No allergies. HIV negative. Packs checked, indicators changed, **baseline count correct and on the board**. Chlorhexidine-alcohol prep **dry**. #1 Vicryl on a round-bodied needle available for the uterus and sheath. Double gloved."

Model presentation (theatre-based viva):

> "For this caesarean section I scrubbed with chlorhexidine 4% for 3 minutes, gowned and double-gloved with the closed technique. The skin was prepared with chlorhexidine-alcohol from the xiphisternum to mid-thigh and allowed to dry. After a Pfannenstiel incision, the sheath was held with Kocher's and opened with Mayo scissors. The peritoneum was opened with Metzenbaum scissors, and a Doyen retractor was placed over the bladder. After delivery, the uterine angles were secured with Green-Armytage forceps and the uterus closed in two layers with #1 Vicryl on a round-bodied needle. The sheath was closed with continuous #1 Vicryl and the skin with 3-0 subcuticular Monocryl. Counts were correct at the uterus, sheath and skin."

The post-procedure checklist (end of every case):

1. **Final count correct** and recorded (Sign out).
2. **Specimens labelled** (e.g. tubes from a tubal ligation, if sent).
3. **Sharps in the safety box** by the user; instruments into **0.5% chlorine for 10 minutes**.
4. **Dressing** on, with a plan for removing it at 24–48 h.
5. **Operation note** records the sutures used for each layer and **when skin sutures come out** (day 5 Pfannenstiel; day 7–10 midline).
6. Any **sharps injury** reported and managed (PEP within 72 h, ideally within 2 h).
7. Significant **sterility breaches** and any **count discrepancy** documented and reported.

10. OSCE Checklist

Station: "Scrub, gown and glove for a caesarean section" (7–10 minutes)

Step	Done well when	Critical?
Cap covers all hair; mask covers nose and mouth; eye protection; apron	No hair visible; mask ties correct	No
Removes jewellery and watch; nails short; checks hands for cuts	States would not scrub with open lesions	Yes
Opens gown and glove packs aseptically; checks indicator and expiry	Names the checks aloud	Yes
Turns on water with elbow or foot; wets hands and forearms	–	No

Step	Done well when	Critical?
Cleans under the nails (first scrub); no brush on the skin	–	No
Antiseptic soap; systematic scrub of fingers (4 sides), web spaces, palms, backs, wrists, forearms to above the elbow	Systematic order	Yes
Scrubs for 2–5 minutes (states the time)	–	No
Keeps **hands above elbows** throughout	Never drops hands	Yes
Rinses fingertips to elbows, one direction	–	No
Enters theatre backwards, hands up and away	–	No
Dries with a sterile towel, one end per arm, fingertips to elbow, no going back	–	Yes
Picks up the gown by the inside neckline; lets it unfold away from the body	Does not touch the outside	Yes
Arms into sleeves; **hands stay inside the cuffs**	–	Yes
Asks the circulator to tie at the back	–	No
Closed gloving: glove palm-to-palm, fingers toward the elbow, rim over the cuff, pushes the hand through	–	Yes
Knitted cuffs completely covered; second glove done correctly	–	Yes
Wrap-around tie passed via the card/tag; turns; ties at the side	–	No
Keeps hands in the sterile zone (front, above the waist); never in the axillae	–	Yes
Identifies and corrects any breach (verbalises "I would change…")	Honest about breaches	Yes
States counts and the WHO checklist "time out" before incision	Antibiotic within 60 min	No

Station: "Identify these instruments and sutures": name, class, features, how to hold it, use at CS, one other use, precautions. For sutures: read the **packet** (material, absorbable?, monofilament or braided?, gauge, needle type and shape, length), then state **which layer** and **why**.

ll. Key Points

High-yield points

- **If in doubt, it is contaminated.** Never close on an incorrect count.
- Doyen = bladder; Green-Armytage = uterine angles; Kocher's/Allis = sheath; Babcock = tube.
- Straight Mayo cuts sutures; **never Metzenbaum**.
- More zeros = thinner: **#1** uterus and sheath, **2-0** fat/peritoneum, **3-0** skin.
- Round-bodied for soft tissue; reverse cutting for skin.
- Monofilament: less infection, more throws. Braided: easy knots, wicks bacteria.
- Scrub **2–5 min**, no brush on skin, hands above elbows.
- Chlorhexidine-alcohol prep from the incision outwards, **dry for 2–3 min** before diathermy; **clip, never shave**.
- Antibiotic within **60 min before incision**; counts at baseline, uterus, peritoneum/sheath, skin.
- Pink serous discharge day 5–10 = impending **burst abdomen**.
- Swab pus **before** antibiotics; local SSI organisms are often ceftriaxone-resistant.

Numbers & doses

Item	Value
Surgical scrub with antiseptic soap	**2–5 min** (chlorhexidine 4% or povidone-iodine 7.5%); longer gives no benefit
Alcohol surgical hand rub	Keep hands wet for the product's contact time (typically **about 1.5–3 min**); plain soap wash first
Scrub area	Fingers to about **5 cm above the elbow**; hands above elbows
Skin prep	**2% chlorhexidine in 70% alcohol** (WHO first choice), about **30 s** of friction; or 10% povidone-iodine
Prep drying time	About **2–3 min** before diathermy (fire risk)
Prep area	**Xiphisternum to mid-thigh**, flank to flank
Distance of unscrubbed staff from the field	At least **30 cm**
Antibiotic prophylaxis	Within **60 min before incision** (confirmed at Time out)
Counts	**Baseline**, before closing the **uterus**, before closing the **peritoneum/sheath**, at **skin**
Uterus and sheath suture	**#1** (Vicryl or chromic catgut for uterus; Vicryl or PDS for sheath)
Subcutaneous / peritoneum	**2-0**; close fat only if **more than 2 cm**
Skin	**3-0** subcuticular (Monocryl/Vicryl) or 2-0/3-0 nylon interrupted
Sheath bites	About **1 cm from the edge, 1 cm apart**
Vicryl strength	About **75% at 2 weeks, 50% at 3 weeks**; absorbed **56–70 days**
PDS	Useful strength about **6 weeks**; absorbed about **6 months**
Plain catgut / chromic catgut strength	About **7–10 days** / about **2–3 weeks**
Knot throws	About **5–6** for nylon, **3–4** for braided
Suture tails	**3–5 mm** braided; **5–10 mm** monofilament
Skin suture removal	**Day 5** Pfannenstiel; **day 7–10** midline
Catheter after bladder repair	**7–14 days**
Instrument decontamination	**0.5% chlorine for 10 min**
Autoclave	**121 °C, 106 kPa, 20 min unwrapped / 30 min wrapped**
Chlorhexidine anaphylaxis	**Adrenaline 0.5 mg IM** (0.5 mL of 1:1000)
Sharps injury PEP	TLD for **28 days**, ideally within **2 h**, never after **72 h**

Exam traps

- **"Cut the suture with the Metzenbaum."** No. Metzenbaum is for delicate dissection; **straight Mayo** scissors cut sutures.
- **"Use Allis to hold the fallopian tube."** No. Allis has teeth; use the **atraumatic Babcock** for tubular structures.
- **"2-0 is thicker than 1."** No. More zeros means **thinner**: #1 > 0 > 2-0 > 3-0.
- **"Shave the abdomen the night before."** No. Shaving increases SSI; **clip** only if needed, just before surgery.
- **"A 10-minute scrub with a brush is best."** No. **2–5 minutes**, no brush on the skin; longer damages skin without benefit.

- **"Start diathermy straight after the alcohol prep."** No. Wait for it to **dry** (2–3 min): wet or pooled alcohol burns.
- **"Nudge the drape towards the incision to adjust it."** No. A drape can only be moved **outwards**; otherwise discard it.
- **"Use silk or nylon to repair the bladder."** No. Non-absorbable sutures in the bladder cause **stones**; use absorbable 2-0/3-0 in two layers.

Questions seniors ask

Viva questions

1. **Why is a round-bodied needle used on the uterus?** It spreads rather than cuts the fibres, leaving a small hole that seals. This means less bleeding and a weaker tendency to tear.
2. **Why is plain catgut used for Pomeroy tubal ligation?** It absorbs quickly, so the cut ends of the tube separate and fibrose apart, which reduces recanalisation.
3. **Why should braided non-absorbable sutures be avoided in contaminated wounds?** Capillarity harbours bacteria in the braid, causing persistent infection and stitch sinuses.
4. **Why is shaving discouraged before surgery?** It causes micro-abrasions that become colonised, and it increases SSI. Clip if hair must be removed.
5. **What is the sterile area of a gown?** The front from chest to waist or table level, and the sleeves from 5 cm above the elbow to the cuff.
6. **How long should a surgical scrub take?** 2–5 minutes with antiseptic soap (per product). Longer gives no benefit.
7. **What is the advantage of chlorhexidine over povidone-iodine for skin prep?** Chlorhexidine-alcohol acts faster, **persists (residual activity)**, is not inactivated by blood, and reduces SSI more (WHO).
8. **When must counts be done at CS?** Before starting, before closing the uterus, before closing the peritoneum or sheath, and at skin closure.
9. **What is the difference between Allis and Babcock?** Allis has **fine interlocking teeth** (traumatic, for tough tissue or tissue to be removed). Babcock has **atraumatic fenestrated rounded tips** (for tubular structures such as the tube or bowel).
10. **What does the number "2-0" mean?** A USP gauge. It is thinner than 0 and thicker than 3-0. More zeros means thinner.

In theatre and on the ward round

1. **What do you check on a sterile pack before opening it?** The **indicator tape has changed colour**, the pack is **dry and intact**, it is **in date**, and it has not been dropped.
2. **The count is one swab short at sheath closure. What happens?** Stop closure, recount, the surgeon explores the cavity, the circulator searches the floor and bins; if still missing, **X-ray** before she leaves theatre, and document.
3. **Why a round-bodied needle for the uterus?** It **spreads** the fibres rather than cutting, leaving a small hole that seals, with less bleeding.
4. **Why PDS rather than Vicryl for the sheath in an obese or infected woman?** It is a **monofilament** (less infection) that keeps strength for about **6 weeks**, while fascia heals slowly.
5. **Pink serous discharge from a midline wound on day 7. What worries you? Fascial dehiscence** (impending burst abdomen). Examine, cover with moist sterile packs if it gapes, and call the surgeon for **re-closure in theatre**.
6. **What are the three pauses of the WHO Surgical Safety Checklist? Sign in** (before anaesthesia), **Time out** (before incision: team, patient, procedure, category, antibiotic given), **Sign out** (before leaving theatre: counts, specimens, concerns).

7. **You puncture your glove on the suture needle. What do you do?** Say so, step back and have the **gloves changed**. If the skin was pierced, hand over or finish the critical step safely, then do **first aid** (let it bleed, wash with soap and water), **report**, and get assessed for **PEP within hours** (ideally within 2 h).

Mnemonics

- **"Doyen Defends the Detrusor"**: the Doyen retracts the bladder.
- **"Green-Armytage for the Gaping Uterine Angles"**.
- **"Babcock Babies the tube"** (atraumatic). **"Allis Attacks"** (teeth).
- **"Metzenbaum for Membranes, Mayo for Muscle and sutures"** (straight Mayo = suture scissors).
- **"More zeros, more zero thickness"**: 4-0 is thinner than 2-0.
- **Suture choice "LCRA"**: **L**ength of support needed, **C**ontamination, **R**emoval needed?, **A**vailability.
- **Scrub order "Fingers first, elbows last, hands up, never back."**
- **Counts at "U-P-S"**: **U**terus, **P**eritoneum/sheath, **S**kin (plus the baseline).
- **Gown sterile zone "Chest to table, sleeves to elbow"**.

OSCE tips

- For an instrument: **name, class, features, grip, use at CS, other use, precaution**.
- For a suture packet: **material, absorbable?, mono/braided, gauge, needle type and shape, length → layer → why**.
- Talk through each step of the scrub aloud and state the time.

> **EXAM FAVOURITE**
> Examiners love **"What is wrong here?"** stations: a student who has not removed a ring, who drops their hands, touches their mask, or folds their arms. Talk through what you are doing, and if you contaminate, **say so and correct it**. Honesty about a breach scores marks. Hiding it fails the station.

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SELF-TEST

Flashcards

25 cards. Cover the right-hand column and test yourself.

#	Question	Answer
1	Doyen retractor: features and use at CS.	A **single broad curved blade** on a handle. It is placed over the pubis to **retract and protect the bladder** and expose the lower segment. It is removed before delivering the head.
2	Green-Armytage forceps: features and use.	Large, long **haemostatic clamps with broad serrated jaws**. They grip the **bleeding edges and angles of the uterine incision** at CS, reducing blood loss and holding the edges for closure.
3	Allis vs Babcock forceps.	**Allis**: fine **interlocking teeth**, traumatic. For tough tissue (sheath, fascia) or tissue that will be removed. **Babcock**: **rounded fenestrated atraumatic tips**. For **tubular structures**: fallopian tube (tubal ligation), bowel, ureter.
4	Kocher's forceps: features and use at CS.	A heavy straight artery forceps with **1×2 teeth at the tip**. It grips and lifts the **rectus sheath** edges so the sheath can be dissected off the rectus muscles (Pfannenstiel).
5	Mayo vs Metzenbaum scissors.	**Mayo**: short, heavy. Curved Mayo cuts **tough tissue (rectus sheath)**; straight Mayo cuts **sutures**. **Metzenbaum**: long handles, fine blades, for **delicate dissection** (peritoneum, uterovesical fold). **Never cut sutures** with them.
6	Scalpel handles and their blades.	**No. 3 handle**: small blades **10, 11, 15**. **No. 4 handle**: large blades **20–24** (No. 22 is common for CS skin). Load and remove blades with a **needle holder**, never with fingers.
7	Poole vs Yankauer sucker.	**Poole**: straight tube with a **perforated guard**, for the **peritoneal cavity** (omentum does not block it). **Yankauer**: rigid curved tube with a bulbous tip, for superficial suction and the **mouth/airway** (gently for babies).
8	Three axes of suture classification.	1. **Absorbable vs non-absorbable** 2. **Monofilament vs multifilament (braided)** 3. **Natural** (catgut, silk) **vs synthetic** (Vicryl, PDS, Monocryl, nylon, Prolene)
9	Polyglactin 910 (Vicryl): type, strength retention, absorption, CS uses.	Synthetic, **braided, absorbable** (hydrolysis). About **75% strength at 2 weeks, 50% at 3 weeks**; absorbed in **56–70 days**. Used for the **uterus and sheath (#1)**, bladder (2-0/3-0), fat (2-0) and subcuticular skin (3-0).
10	Chromic vs plain catgut.	Both are natural and absorbable (enzymatic, unpredictable, faster with infection). **Plain**: strength **7–10 days**, absorbed in about **70 days**. Used for tubal ligation and subcutaneous tissue. **Chromic**: chromium-treated, strength about **2–3 weeks**, absorbed in about **90 days**. Used for the uterus in many Ugandan theatres.
11	PDS (polydioxanone): why use it for the rectus sheath?	It is a **monofilament absorbable** suture that keeps useful strength for about **6 weeks** (absorbed in about 6 months). Fascia heals slowly, and the monofilament resists infection. Good for obese, infected or midline wounds.
12	Monofilament vs braided: pros and cons.	**Braided**: easy handling, secure knots, but **capillarity** harbours bacteria (infection, sinuses). **Monofilament**: smooth, less drag, **less infection**, but **memory**, so knots slip and need more throws (about 5–6 for nylon).

#	Question	Answer
13	What does USP gauge 3-0 mean relative to 0 and #1?	More zeros means **thinner**. #1 > 0 > 2-0 > 3-0 > 4-0. At CS: **#1** for uterus and sheath, **2-0** for fat and peritoneum, **3-0** for skin.
14	Round-bodied vs cutting vs reverse-cutting needles.	**Round-bodied (taper)**: spreads the tissue. For **uterus, peritoneum, bowel, bladder**. **Cutting**: edge on the **inner** curve. For skin. **Reverse cutting**: edge on the **outer** curve. Less cut-out; for **skin and tough fascia**.
15	WHO surgical hand antisepsis with antiseptic soap.	Remove jewellery; short nails; clean nails under running water (first scrub; no brush on the skin). Use **chlorhexidine 4% or povidone-iodine 7.5%** for **2–5 min**, covering fingers (4 sides), hands and forearms to above the elbow. **Hands above elbows**; rinse fingertips to elbow; dry with a sterile towel.
16	What is the sterile zone of a surgical gown?	The **front from chest to waist / table level** and the **sleeves from about 5 cm above the elbow to the cuff**. The back, neckline, shoulders, **axillae** and everything below the waist are NOT sterile.
17	Describe closed gloving.	Hands stay **inside the sleeves**. Place the glove **palm to palm with its fingers pointing to the elbow**, grasp the rim through the sleeve, flip it **over the cuff**, then push the hand through. The knitted cuff must be fully covered.
18	Best skin prep for CS and the key safety point.	**2% chlorhexidine gluconate in 70% alcohol** (WHO first choice), or povidone-iodine. Paint from the incision outwards, xiphisternum to mid-thigh and flank to flank. **Let it dry fully**: wet alcohol plus diathermy means **fire and burns**. Do not shave.
19	When are counts done at CS?	**Baseline** before incision, then **before closing the uterus**, **before closing the peritoneum/sheath**, and **at skin closure**. Items added during the case are counted in. **Never close on an incorrect count.**
20	Why is plain catgut chosen for Pomeroy tubal ligation, and why no non-absorbable sutures in the bladder?	**Plain catgut** absorbs quickly, so the cut tubal ends separate and fibrose apart (less recanalisation). **Non-absorbable sutures in the bladder** act as a nidus for **bladder stones**. Use 2-0/3-0 Vicryl in 2 layers.
21	What must the scrubbed assistant confirm before the first incision at CS (pre-incision workup)?	**Packs**: indicator changed, dry, intact, in date. **Instruments and sutures** present (#1 round-bodied for uterus and sheath). **Suction and diathermy** working, plate on dry skin. **Baseline count** on the board. **Prep dry. Time out** done: patient, procedure, category, allergies, **antibiotic given within 60 min**, blood available, neonatal resuscitator present.
22	The count is one swab short before sheath closure. What are the steps?	1. **Stop closing** 2. Scrub nurse and circulator **recount** 3. Surgeon **explores** the gutters, pouch of Douglas, under the uterus and the subhepatic space 4. Circulator searches the **floor, bins, drapes and linen** 5. Still missing: **X-ray** (radio-opaque marker) before she leaves theatre 6. **Document** and report as an incident
23	Post-op wound: how do you tell a normal wound, an SSI and an impending dehiscence apart?	**Normal**: dry, edges together, pain improving after day 2. **SSI** (day 4–7): spreading redness, warmth, induration, pus, fever. Swab, open and drain, antibiotics if cellulitis or systemic signs. **Fascial dehiscence** (day 5–10): **pink serous discharge**, then gaping, bowel visible. Moist sterile packs, IV fluids and antibiotics, **re-closure in theatre**.

#	Question	Answer
24	When should a pus swab be taken from an infected CS wound, and why does it matter in Uganda?	**Before starting antibiotics.** Ugandan SSI isolates are often **ESBL-producing *Klebsiella/ E. coli*** (ceftriaxone-resistant) or **MRSA**, so an empirical regimen may not work. Start sensible empirical treatment if the woman is systemically unwell, then **change to the culture result**, and alert IPC if there is a cluster.
25	Which pre-op patient factors change your choice of sheath and skin closure, and how?	**Obesity, diabetes, steroids, anaemia, HIV with a low CD4, infection/chorioamnionitis** all impair healing. Consider **#1 PDS (monofilament, long-lasting) for the sheath**, **close fat if >2 cm**, prefer **monofilament** over braided in a contaminated field, and use **interrupted nylon** for skin where infection is likely (it can be partly removed to drain). Check and control **glucose** and correct **anaemia**.

SELF-TEST

Single best answer questions

15 questions. The answers and explanations follow at the end.

1. At the end of a caesarean section, the patient has consented to tubal ligation. The surgeon asks you to pass an instrument to lift the mid-portion of the right fallopian tube without crushing it. Which instrument do you pass?

A Allis tissue forceps
B Babcock forceps
C Kocher's forceps
D Green-Armytage forceps
E Towel clip

2. You are assisting at a caesarean section for an HIV-positive woman. While retracting, you notice a small tear in your outer glove. You have not been pricked and the inner glove is intact. What is the MOST appropriate action?

A Continue; the inner glove is intact so there is no breach
B Wipe the torn glove with an alcohol swab and continue
C Pull another glove on over the torn one without stepping back
D Tell the surgeon, step back from the field and have the outer glove changed by the scrub nurse
E Finish the operation and change the gloves before the next case

3. A 34-year-old with a BMI of 42 has an emergency CS for chorioamnionitis with foul liquor. The surgeon is about to close the rectus sheath and asks for your suggestion. Which suture is MOST appropriate? Advanced

A #1 polydioxanone (PDS), continuous
B 2-0 plain catgut, continuous
C #1 silk, interrupted
D 3-0 Vicryl Rapide, continuous
E 2-0 poliglecaprone (Monocryl), subcuticular

4. You are preparing the abdominal skin of a woman for elective CS. Diathermy will be used. According to WHO guidance, which is the BEST approach?

A Shave the pubic hair on the ward the night before, then paint with povidone-iodine
B Aqueous chlorhexidine, then diathermy immediately while the skin is still wet
C 70% alcohol alone, applied from the periphery towards the incision
D Soap and water wash in theatre, then drape
E 2% chlorhexidine in 70% alcohol from the incision site outwards, allowed to dry completely before draping and diathermy

5. At the closing count before the rectus sheath is closed, the scrub nurse reports that one abdominal swab is missing. The surgeon is sure she has checked the cavity. What is the MOST appropriate next step?

A Close the sheath and request an abdominal X-ray tomorrow on the ward
B Document the discrepancy and close, as swabs are often miscounted
C Stop closure, recount, systematically explore the cavity and search the floor, drapes and bins, and obtain an X-ray before she leaves theatre if it is still missing
D Close and give a 7-day course of antibiotics in case a swab was left behind
E Ask the circulating nurse to change the count on the board to match

6. A junior clerk is scrubbing for her first CS at Mengo Hospital with chlorhexidine 4% soap. Which technique is CORRECT according to the WHO?

A Scrub hands and forearms for 2–5 minutes with hands held above the elbows, and rinse from fingertips to elbows

B Scrub with a hard brush over the skin for 10 minutes to remove all bacteria

C Keep the hands below the elbows so water drains from the arms to the fingertips

D Rinse from the elbows down to the fingertips so the hands are rinsed last

E A 20–30 second hygienic hand rub is sufficient before gowning

7. After delivery of the baby and placenta at CS, the surgeon asks for the suture and needle to close the first layer of the uterus. Which needle is MOST appropriate?

A Reverse-cutting ⅜-circle needle

B Straight cutting (hand-held) needle

C Cutting ½-circle needle

D Round-bodied (taper) ½-circle needle

E Blunt-point straight needle

8. For a modified Pomeroy tubal ligation at CS, the surgeon asks for a suture that will absorb quickly so that the cut ends of the tube separate. Which do you pass?

A 2-0 nylon

B #1 plain catgut

C #1 silk

D #1 polydioxanone (PDS)

E #1 polypropylene (Prolene)

9. After gowning and gloving, a student waits for the surgeon with his gloved hands tucked into his armpits 'to keep them clean'. The scrub nurse notices. What is the CORRECT response?

A This is acceptable, as the gown covers the axillae

B It is acceptable as long as the gloves did not touch bare skin

C Spray the gloves with alcohol hand rub and proceed

D He must go back and re-scrub for 10 minutes before regowning

E The axillae are not sterile, so his gloves (and gown, if the sleeves are soiled) are contaminated and must be changed

10. During a difficult repeat CS, the bladder dome is inadvertently opened (a 2 cm cystotomy). The surgeon will repair it. Which suture is MOST appropriate? Advanced

A #1 silk, interrupted, full thickness

B 2-0 nylon, continuous, through the mucosa

C 2-0 or 3-0 Vicryl in two layers

D #1 polypropylene, one layer

E No repair needed; leave a catheter for 24 hours

11. On day 7 after a midline laparotomy for ruptured uterus, a 30-year-old notices a large amount of pink serous fluid soaking her dressing. On inspection, the wound has separated and a loop of bowel is visible. She is anaemic (Hb 8.5 g/dL). What is the MOST appropriate immediate management? Advanced

A Push the bowel back and strap the abdomen tightly with an abdominal binder

B Cover the wound with sterile saline-soaked packs, reassure her, start IV fluids and antibiotics, and prepare for return to theatre for mass closure

C Apply a dry gauze dressing and continue oral antibiotics

D Leave the wound to heal by secondary intention with daily dressings

E Discharge with a follow-up appointment in 1 week

12. Immediately after delivery of the placenta at CS, the lateral angles of the uterine incision are bleeding briskly. Which instrument should you pass to the surgeon FIRST?

A Doyen retractor

B Metzenbaum scissors

C Babcock forceps

D Towel clip

E Green-Armytage forceps

13. While draping for a CS, a student places the lateral towel drape too far from the planned incision, leaving a wide strip of prepared skin exposed. She slides the drape inwards towards the incision line. What is the CORRECT management?

A Discard that drape and place a fresh sterile drape, because a drape must never be moved towards the incision

B It is acceptable, as the skin has already been prepared

C Re-paint the skin under the drape with chlorhexidine and continue

D Secure it in its new position with extra towel clips

E Place the large fenestrated drape over it and proceed

14. The surgeon closing the skin with 3-0 nylon interrupted sutures asks the student to put at least 5 throws on each knot. Which property of nylon explains this?

A It is absorbable, so the knot may dissolve early

B It is braided, so the knot slips along the braid

C It is a natural material with high tissue reactivity

D It is a monofilament with high memory and low friction, so knots tend to slip

E It has high capillarity and harbours bacteria

15. Needle-stick injuries in the Mengo theatre have increased. Which intraoperative practice will MOST effectively reduce sharps injuries among the scrubbed team?

A Passing the scalpel hand to hand with the blade pointing towards the surgeon

B Recapping used needles with two hands before placing them on the trolley

C Passing sharps through a neutral zone (kidney dish), announcing 'sharp', and using instruments rather than fingers to handle needles and blades

D Removing scalpel blades with the fingers to save time

E Leaving used needles on the patient's drape for the scrub nurse to collect later

Answers & explanations

1. B · Babcock forceps

The **Babcock** has rounded, fenestrated, **atraumatic** tips designed for tubular structures such as the tube and bowel. **Allis** and **Kocher's** have teeth, so they crush and tear delicate tissue. **Green-Armytage** clamps are for the uterine incision edges. A **towel clip** has sharp points for securing drapes.

2. D · Tell the surgeon, step back from the field and have the outer glove changed by the scrub nurse

A torn glove is a **breach of sterility** and exposes you to blood. **Announce it, step back and have the glove changed** (assisted gloving) (**D**). **A** and **E** leave a contaminated, non-barrier surface in the wound. **B**: alcohol does not re-sterilise a torn glove. **C**: gloving over a torn glove traps contamination and is not how a breach is corrected. Double gloving is recommended in high-prevalence settings precisely so that this situation does not expose your skin.

3. A · #1 polydioxanone (PDS), continuous

Fascia heals slowly, and obesity and infection raise the risk of dehiscence, so she needs a **strong, long-lasting, monofilament** suture. **#1 PDS** keeps strength for about 6 weeks and resists infection (**A**). **B**: plain catgut loses strength in 7–10 days and is too thin. **C**: braided non-absorbable silk in an infected wound causes **stitch sinuses and persistent infection**. **D**: Vicryl Rapide loses half its strength in 5 days (it is for the perineum). **E**: 2-0 Monocryl is a skin (subcuticular) suture and far too weak for the sheath.

4. E · 2% chlorhexidine in 70% alcohol from the incision site outwards, allowed to dry completely before draping and diathermy

The WHO SSI guideline strongly recommends **alcohol-based chlorhexidine** for skin prep. It must be applied from **clean (incision) to dirty (periphery)** and allowed to **dry** before diathermy, to avoid fires and burns (**E**). **A**: **shaving increases SSI**; clip only if needed, just before surgery. **B**: diathermy on wet prep risks fire, and the antiseptic hasn't had time to work. **C**: alcohol alone has no residual activity, and periphery-to-centre drags bacteria into the incision site. **D**: soap is not an antiseptic prep.

5. C · Stop closure, recount, systematically explore the cavity and search the floor, drapes and bins, and obtain an X-ray before she leaves theatre if it is still missing

An incorrect count means **stop and search** (**C**): recount, explore the gutters, pelvis and pouches, search outside the field, and use an X-ray (radio-opaque markers) if it is still unresolved. A retained swab causes sepsis, abscess, fistula and obstruction. **A**, **B** and **D** accept a possible retained foreign body and delay retrieval, which would need a second laparotomy. **E** is falsification of records and unethical.

6. A · Scrub hands and forearms for 2–5 minutes with hands held above the elbows, and rinse from fingertips to elbows

WHO: antiseptic scrub for **2–5 minutes**, **hands above elbows**, and **rinse fingertips to elbows** so water flows from clean to less clean (**A**). **B**: long brush scrubs damage the skin and increase bacterial shedding, with no added benefit. **C** and **D** let dirty water from the elbow run onto the clean hands. **E**: a hygienic 20–30 s hand rub is for ward care, not surgical antisepsis.

7. D · Round-bodied (taper) ½-circle needle

Uterine muscle is soft and vascular. A **round-bodied (taper) needle** spreads the fibres rather than cutting them, so the hole seals, bleeding is minimal and it is less likely to tear. A ½ circle suits the deep, confined field (**D**). **A** and **C**: cutting needles slice the myometrium, causing bleeding and tears; they are for skin. **B** and **E**: straight needles are unsuitable in a deep cavity and raise the needle-stick risk.

8. B · #1 plain catgut

The Pomeroy technique relies on **rapidly absorbed plain catgut**. The ligature dissolves and the cut ends of the tube pull apart and fibrose separately, which reduces recanalisation (**B**). **A**, **C** and **E** are non-absorbable and would hold the ends together, favouring recanalisation or fistula. **D**: PDS persists for months.

9. E · The axillae are not sterile, so his gloves (and gown, if the sleeves are soiled) are contaminated and must be changed

The **axillae are outside the sterile zone** (the gown is sterile only on the front from chest to table level and on the sleeves to above the elbow). Gloves tucked there are contaminated and **must be changed**, along with the gown if it is compromised (**E**). **A** and **B** misunderstand the sterile zone. **C**: alcohol does not re-sterilise gloves. **D**: a 10-minute re-scrub is unnecessary and not the WHO standard. Changing the contaminated gloves and gown is what is needed.

10. C · 2-0 or 3-0 Vicryl in two layers

Bladder repair uses an **absorbable** suture (2-0/3-0 Vicryl or chromic) in **two layers** with a catheter for 7–14 days (**C**). **A, B and D**: non-absorbable sutures in the bladder become a nidus for **stones** and infection. **E**: an unrepaired cystotomy leaks urine into the peritoneum, and 24 h of drainage is inadequate.

11. B · Cover the wound with sterile saline-soaked packs, reassure her, start IV fluids and antibiotics, and prepare for return to theatre for mass closure

This is **burst abdomen (fascial dehiscence)**. The typical warning is a 'pink serous' discharge around day 5–10. Risk factors here are the midline incision, anaemia, sepsis and emergency surgery. **Cover the bowel with warm moist sterile packs, give IV fluids and antibiotics, and return to theatre for mass closure** (e.g. #1 loop nylon or PDS) (**B**). **A**: pushing contaminated bowel back and strapping without repair risks injury and evisceration. **C** and **D** are inadequate for exposed bowel. **E** is dangerous.

12. E · Green-Armytage forceps

Green-Armytage forceps are designed to grasp the **edges and angles of the uterine incision**, controlling bleeding and presenting the edges for suturing (**E**). **A**: the Doyen retracts the bladder; it is not a haemostat. **B**: Metzenbaum scissors dissect. **C**: the Babcock is atraumatic but too small and not haemostatic for the uterine edge. **D**: towel clips secure drapes.

13. A · Discard that drape and place a fresh sterile drape, because a drape must never be moved towards the incision

Once placed, a drape may only be moved **away from** the incision. Sliding it inwards drags skin organisms from the periphery (less clean) into the operative site. **Discard it and use a fresh drape** (**A**). **B**: prepared skin is not sterile, just less contaminated. **C**, **D** and **E** leave the contaminated drape edge at the wound.

14. D · It is a monofilament with high memory and low friction, so knots tend to slip

Monofilament synthetics (nylon, Prolene) have **memory** (they spring back to their packaged shape) and a **smooth surface**. Knots loosen unless more throws are placed (**D**). **A**: nylon is non-absorbable. **B**: nylon is monofilament; braided sutures actually hold knots better. **C**: nylon is synthetic and relatively inert. **E**: capillarity is a feature of braided sutures.

15. C · Passing sharps through a neutral zone (kidney dish), announcing 'sharp', and using instruments rather than fingers to handle needles and blades

The **neutral zone (hands-free) technique**, announcing sharps, and using needle holders or forceps to handle needles and blades are proven ways to reduce sharps injuries (**C**). **A** and **D** put hands directly in the path of the blade. **B**: **never recap by hand**; two-handed recapping is a classic cause of needle-sticks. **E**: loose sharps on drapes injure the team and can be lost in the wound.

SELF-TEST

Clinical cases

Work through each step before reading the model answer.

Case 1: Your first scrub at Mengo theatre

You are asked to scrub in as second assistant for an elective CS (two previous scars) at Mengo Hospital. The patient is a 33-year-old P2+0, HIV-negative, with no allergies. The scrub nurse has laid out the CS set. You are wearing a ring and a wristwatch.

1. What do you do before approaching the scrub sink?

- Cap covering all hair, **mask covering nose and mouth**, eye protection, **plastic apron**, theatre shoes.
- **Remove the ring and watch** (and bracelets). Keep nails short, no polish.
- Check hands for **cuts** (cover or don't scrub).
- Check that the gown and glove packs are **sterile** (indicator changed, dry, intact, in date) and have them opened onto a separate surface.

2. Describe your surgical hand antisepsis.

Turn on the water with the elbow or foot. Wet hands and forearms. Clean **under the nails** under running water (first scrub). Apply **chlorhexidine 4%** and scrub systematically: each finger's 4 sides, web spaces, palms, backs, wrists, and forearms to 5 cm above the elbow, for **2–5 minutes**. **Hands above elbows** throughout. **Rinse fingertips to elbows** in one direction. Enter theatre backwards with hands up. Dry with a sterile towel, one end per arm, fingertips to elbow, never going back.

3. While gowning, your bare knuckle touches the outside front of the gown. What do you do?

The **outside of the gown is now contaminated**. Say so. Discard that gown and put on a new one. Your hands are still scrubbed, so you do not need to re-scrub unless you touched something unsterile. With **closed gloving**, your hands should never come out of the cuffs, which prevents exactly this problem.

4. Describe closed gloving and how you secure the gown.

With hands inside the sleeves, place the glove **palm to palm, fingers towards the elbow**. Grasp the lower rim through the sleeve, flip the upper rim **over the cuff** with the other (covered) hand, then push the hand through. Repeat with the other hand. **Double glove** (recommended for CS in Uganda). Pass the wrap-around tie card to the circulator, turn, take the tie and fasten it at your side. Keep hands in front, above the waist, and never in the axillae.

5. The surgeon asks you to prepare the skin. How?

Use **sponge-holding forceps** with gauze soaked in **2% chlorhexidine in 70% alcohol**. Start at the incision site with back-and-forth friction for about 30 s and work **outwards**, covering the **xiphisternum to mid-thigh** and **flank to flank**. Use fresh swabs, never returning to the centre. Clean the umbilicus. Let it **dry completely (2–3 min)** before draping and diathermy, and avoid pooling under the patient.

6. What must happen before the skin incision?

Drapes are placed correctly (four towels, towel clips, fenestrated drape; never moved inwards). **Baseline count** of swabs, needles and instruments aloud by the scrub nurse and circulator, recorded on the board. **WHO checklist Time Out**: team introductions; patient, procedure and category confirmed; **antibiotic given within 60 min**; blood available; anticipated difficulties (adhesions with two previous scars). Confirm the **fetal heart** is present.

Take-away: Aseptic technique is a sequence of small disciplined steps: no jewellery, correct scrub, closed gloving, a dry chlorhexidine-alcohol prep, correct draping, baseline count and Time Out. Admit and correct every breach immediately.

Case 2: Anticipate the surgeon: instruments and sutures layer by layer

You are first assistant at a Category 2 CS for fetal distress in a 26-year-old P1+0 at an HC IV. The medical officer will perform a Pfannenstiel incision. The scrub nurse is new and asks you to help her anticipate each instrument and suture.

1. Skin to peritoneum: which instruments are needed at each step?

- **Skin**: scalpel (No. 4 handle, No. 22 blade), toothed dissecting forceps, dry swabs, and artery forceps or diathermy for bleeders.
- **Fat**: fingers to separate, Langenbeck retractors.
- **Sheath**: scalpel to nick, **curved Mayo scissors** to extend, **Kocher's** to lift the sheath edges while it is freed from the recti.
- **Peritoneum**: two **artery forceps** (or non-toothed forceps) to tent it, check nothing is caught, open with **Metzenbaum** or fingers, high up (bladder).

2. Uterine incision and delivery: what next?

Doyen retractor over the bladder. Non-toothed forceps and **Metzenbaum** scissors to open the uterovesical fold. A **fresh scalpel** for the uterine incision (a pen grip, stroke by stroke), then **blunt extension** with fingers. Remove the Doyen before delivering the head. Have **Yankauer suction**, 2 cord clamps and cord scissors, and a warm towel for the baby. The circulator draws up **oxytocin** per protocol (**5 IU by slow IV injection** per NICE, or **10 IU IM/slow IV** per WHO; never a rapid IV bolus, which causes hypotension).

3. The placenta is out and the uterine angles are bleeding. What do you pass, and which suture and needle for the uterus?

Green-Armytage forceps for the angles and edges, and a **dry swab** to check the cavity is empty. For closure: **#1 Vicryl (or #1/#2 chromic catgut) on a round-bodied ½-circle needle** (about 40 mm). The first layer is continuous locking, the second imbricating. **Straight Mayo scissors** to cut sutures (tails 3–5 mm for braided), not Metzenbaum.

4. When is the first closing count, and what happens if there is a discrepancy?

The **first closing count is before closing the uterus** (or at the latest, before closing the peritoneum/sheath). If one swab is missing: **stop, recount, explore** (paracolic gutters, behind the uterus, vesico-uterine pouch), search the floor, bins and drapes, and **X-ray** if it is still missing. Never close on an incorrect count, and document it.

5. Sheath, fat and skin: which sutures and why?

- **Sheath**: **#1 Vicryl continuous** (or PDS loop). It needs weeks of strength because fascia heals slowly. Bites 1 cm from the edge and 1 cm apart.
- **Fat**: close with 2-0 plain or Vicryl **only if >2 cm thick**.
- **Skin**: **3-0 Monocryl subcuticular** (cosmetic, no removal) on a reverse-cutting needle, or **2-0/3-0 nylon interrupted** (removal on day 5) if the infection risk is high.
- **Final count** at skin closure, then Sign Out.

Take-away: A good assistant thinks one step ahead. Kocher's and Mayo for the sheath, Metzenbaum for the peritoneum, Doyen for the bladder, Green-Armytage for the uterine angles, and round-bodied #1 for the

uterus. Each suture is chosen for how long the tissue needs support and how contaminated the field is.

Case 3: A discharging wound on day 6

A 29-year-old P3 (BMI 36, Hb 9.0 g/dL) returns on day 6 after an emergency CS in labour. Membranes had been ruptured for 26 hours and she had a fever in labour. The sheath was closed with #1 chromic catgut and the skin with interrupted silk. She reports a painful wound with a pus-like discharge and fever.

1. What history and examination will you focus on?

> **History**: onset and character of the discharge (**pus vs pink serous**, which suggests dehiscence), fever, pain out of proportion (necrotising infection), vomiting or distension, lochia smell, urinary symptoms, breast symptoms, calf pain, and the theatre record (antibiotic timing, duration, sutures). **Examination**: vitals and sepsis screen; wound erythema, induration, fluctuance, gaping, crepitus and skin necrosis; the **integrity of the sheath** (cough impulse, gaping deep layer); uterine tenderness and lochia; breasts, chest and calves.

2. The wound is red and indurated around two silk sutures, with pus from the lateral end; the sheath feels intact. Temp 38.3 °C. What is your diagnosis and management?

> **Superficial incisional SSI** (possibly with stitch abscesses from the braided silk).
>
> 1. **Remove the skin sutures** over the infected area to **open and drain** it; send pus for **culture and sensitivity**.
> 2. Irrigate with saline and apply a **daily** moist dressing; allow healing by secondary intention or delayed closure later.
> 3. **Antibiotics** because of cellulitis and fever: e.g. amoxicillin-clavulanate, or cloxacillin plus metronidazole orally, or IV ceftriaxone plus metronidazole if systemically unwell. Adjust to culture, since MRSA and ceftriaxone-resistant Gram-negatives are common in Kampala.
> 4. Treat anaemia (ferrous sulphate + folic acid), give nutrition advice, and check blood glucose (obesity).
> 5. Look for co-existing **endometritis**.

3. How did the choice of materials and technique contribute, and what could have reduced the risk?

> - **Braided silk** on the skin harbours bacteria (capillarity). A monofilament (nylon) or subcuticular Monocryl is better.
> - **Chromic catgut** absorbs faster and less predictably in infected tissue, so the sheath may weaken. **#1 PDS or Vicryl** is more reliable.
> - Prevention: antibiotic **before incision** (and therapeutic antibiotics for the fever in labour), **vaginal cleansing**, **chlorhexidine-alcohol** skin prep, no shaving, gentle tissue handling, **closing fat >2 cm**, and glove change after vaginal disimpaction.
> - Patient factors: obesity, anaemia, prolonged ROM, chorioamnionitis.

4. Two days later she coughs and a gush of pink fluid soaks the dressing; you see omentum in the wound. What now?

> **Fascial dehiscence (burst abdomen).** Lie her flat and reassure her. **Cover with warm sterile saline-soaked packs**. Nil by mouth, IV fluids, IV antibiotics, analgesia, Hb and cross-match. Inform the senior and **return to theatre** for debridement and **mass closure** (e.g. #1 loop nylon or PDS; consider tension sutures), with a long-term plan for incisional hernia surveillance.

Take-away: Suture material and asepsis directly affect wound outcomes. Avoid braided non-absorbable sutures in contaminated wounds, use a long-lasting monofilament for the sheath in high-risk women, drain SSIs by removing sutures, and recognise pink serous discharge as the warning of dehiscence.

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A study aid, not a protocol. Check doses against the Uganda Clinical Guidelines 2023, your unit's protocol and a senior before you treat a patient.